Machine Learning Group, Leuphana University of Lüneburg

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Exercise 1

Due: Tuesday, April 15, 2025

Task 1

Compute the derivative of the following functions:

(i) $f(\lambda) = 20\lambda^5$

(ii) $g(z) = 2z^3 + 4z^2$

(iii) $h(\phi) = 7\phi + 10$

(iv) $u(\theta) = 4\theta^5 - 2\sqrt{\theta} + \frac{3}{\theta} + 8$

(v) $v(\gamma) = \ln(2 + 3\gamma^2)$

Task 2

Let $f : \mathbb{R} \to \mathbb{R}$ be a function given by:

$$f(x) = \begin{cases} (x+2)^2, & x < 0, \\ 4 - 2x, & x \geq 0. \end{cases}$$

(i) Sketch the plot of $f(x)$ for $x \in [-2, 2]$.

(ii) Compute the area under the curve of function $f(x)$ for $x \in [-2, 2]$.

Task 3

Compute the following matrix products, if possible:

a.

$$\begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

b.

$$\begin{bmatrix} 1 & 2 & 1 & 2 \\ 4 & 1 & -1 & -4 \end{bmatrix} \begin{bmatrix} 0 & 3 \\ 1 & -1 \\ 2 & 1 \\ 5 & 2 \end{bmatrix}$$

Task 4

Determine the inverses of the following matrices, if possible:

a.

$$\begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{bmatrix}$$

b.

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

Task 5

In a café there are 3 different kinds of cake to choose from. An order of 10 pieces of cake consists in specifying the number of pieces of each cake ordered. How many different orders are there for the 10 pieces of cake, if at least one piece of each kind is ordered?

Hint: The ordered pieces of cake are lined up next to each other in such a way that similar pieces of cake are grouped in one (non-empty) block. To mark the boundaries of each block, think of two flags placed in the spaces between the pieces. How many possibilities are there for the placement of these flags?

Task 6

Consider the following random experiments and events, and then compute the required probabilities.

Hint: Remember that an event $E \subset \Omega$, i.e., E is a subset of the sample space; and the sample space Ω contains all possible outcomes of the random experiment (possible values of a random variable). Additionally, remember that, for a discrete random variable X and an event E, we have that: $P(E) = \sum_{x \in E} P(X = x)$.

(i) A fair die is rolled and X is the number that comes on top. Determine the probability of the events: $A = X \text{ is even}$ and $B = X \text{ is divisible by 3}$.

(ii) A shipment with 100 light bulbs contains 15 defective bulbs. One bulb is taken at random. Determine the probability of the events: $A =$ the taken bulb is defective and $B =$ the taken bulb is not defective.

(iii) There are 6 red socks and 8 blue socks in a drawer. In the complete darkness (i.e. at random), you pull two socks out of the drawer (one at a time). What is the probability of the events: (a) two red socks are pulled, (b) two blue socks are pulled, (c) the two socks have different colors, and (d) the two sock have the same color.
Hint: You can treat the colors of the pulled socks as two random variables: X_1 and X_2. In that way, you can resort to the basic rules of probability (marginal, conditional, product, etc.) to express the joint probability $P(X_1, X_2)$ such that you can compute things like $P(X_1 = \text{red}, X_2 = \text{blue})$.

Task 7

The probability that a bus is late is 0.3. If the bus is late, the student S arrives at the lecture on time with probability 0.2. If the bus is not late, S will be on time for the lecture with probability 0.99. What is the probability that S arrives on time for the lecture?

Task 8

The same 100 students take two exams: Statistics and Math. The compiled results are the following:

	S	S^c
M	62	13
M^c	10	15

where S=Passed on Statistics, S^c=Failed on Statistics, M=Passed on Math, and M^c=Failed on Math.

(i) Calculate the following probabilities regarding a randomly chosen student:

(a) Passed on Statistics given that passed on Math.

(b) Passed on Math given that passed on Statistics.

(c) Failed on Math given that passed on Statistics.

(ii) Are the events M and S statistically independent?
Hint: If events A and B are independent, then $P(A|B) = P(A)$ and $P(B|A) = P(B)$.